GSoC 2026 Project Proposal: Physics Guided Machine Learning on Real Lensing Images

Organization: ML4SCI
Mentors: Michael Toomey, Sergei Gleyzer, Ashutosh Ojha, Pranath Reddy
Author: Ishaan Khadse

Table of Contents

1 Introduction and Contributor Information
2 Abstract
3 Technical Details
3.1 Gravitational Lensing Physics and Machine Learning Intersections
3.2 Deficiencies of Purely Data Driven Deep Learning on Real Lensing Images
3.3 Core Methodology I: Lensing Constraints Embedded in Loss Functions
3.4 Core Methodology II: Analytical Mass Profile Priors (SIS and NFW)
3.5 Core Methodology III: Bayesian Neural Networks for Uncertainty Quantification
3.6 Supplementary Methodology I: Equivariant Neural Networks (ENNs)
3.7 Supplementary Methodology II: Differentiable Ray Tracing Integration
3.8 Supplementary Methodology III: Multi Fidelity Learning Framework
3.9 Evaluation Task Results and Algorithmic Approach
4 Project Deliverables
4.1 Objective and Implementation Details
4.2 Timeline for Deliverables
5 Other Information
5.1 Why ML4SCI?
5.2 Academic Background
5.3 Past Experience
5.4 Past Interaction with Mentors
5.5 Commitments
6 Post GSoC Improvements

1 Introduction and Contributor Information

Name: Ishaan Khadse
Location: Mumbai | Goa
Email: ishaan.khadse007@gmail.com
GitHub: github.com/Ishaan-Khadse
LinkedIn: linkedin.com/in/ishaankhadse/
University: Birla Institute of Technology and Science Pilani, Goa Campus

Major: Bachelor of Engineering (B.E.), Mechanical Engineering
Expected Graduation: 2029
I am an engineering undergraduate at BITS Pilani, deeply interested in scientific machine learning, astrophysics and computational physics. What excites me most is working at the intersection of physics and AI, using modern machine learning techniques to better understand complex physical systems.
My academic journey combines the fundamentals of mechanical engineering with hands on exploration of deep learning. Along the way I've supplemented my coursework with certifications in advanced machine learning algorithms, as well as neural networks and reinforcement learning to strengthen both my intuition and technical depth.
Looking ahead, I am particularly interested in using physics informed neural networks to explore fundamental questions in astrophysics, especially in understanding dark matter through gravitational lensing.

2 Abstract

Strong gravitational lensing occurs when the immense gravitational field of a massive foreground object, such as a galaxy or galaxy cluster, drastically warps the fabric of spacetime, acting as a cosmic lens that magnifies and distorts the light from a background source. These distortions manifest as highly complex topological features, including multiple distinct images, extended giant arcs, and complete Einstein rings. Crucially, the precise morphological structure of these lensing artifacts is hypersensitive to the mass distribution of the foreground lens, including the presence of sub galactic dark matter halos (substructures). By meticulously mapping these substructures, the astrophysical community can probe the fundamental nature of dark matter, differentiating between competing cosmological paradigms such as Cold Dark Matter (CDM), Warm Dark Matter (WDM), and Axion like particles.
Historically, deep learning architectures applied to gravitational lensing such as Convolutional Neural Networks (CNNs) within the DeepLense pipeline have achieved remarkable classification accuracies when trained and evaluated on idealized, simulated datasets. However, the transition from simulated environments to real observational data presents a formidable challenge. Real lensing images, captured by instruments such as the Hubble Space Telescope (HST) or the upcoming Euclid and LSST surveys, are deeply contaminated by complex observational systematics. These include heterogeneous noise distributions, Point Spread Function (PSF) blurring, cosmic ray interference, and the luminous baryonic emission of the foreground lens galaxy. Standard data driven CNNs frequently fail to generalize in these environments, often misinterpreting noise artifacts as dark matter substructures and outputting physically impossible, overconfident predictions.
This GSoC 2026 project proposes the development of a comprehensive Physics Informed Neural Network (PINN) framework specifically engineered to operate on real strong gravitational lensing images. Following detailed consultations with my mentor, Pranath Reddy and my astrophysics professor Prasanta Kumar Das, I have designed an architecture that strictly enforces the physical laws of General Relativity within the deep learning optimization process. Prof. Das recommended three foundational pillars for this project: the embedding of

lensing constraints directly into the loss function, the utilization of mass profile priors (Singular Isothermal Sphere and Navarro-Frenk-White models), and the implementation of Bayesian approaches for rigorous uncertainty quantification.
To supplement these strategies and ensure maximum resilience on real data, my proposal introduces three additional state of the art methodologies. First, I will integrate Equivariant Neural Networks (ENNs) to natively preserve the rotational and translational symmetries inherent in dark matter morphologies. Second, I will employ Differentiable Ray Tracing via PyTorch to enable end to end backpropagation through the actual physical rendering process of the lens. Third, I will implement Multi Fidelity Learning to bridge the critical domain gap between abundant Lenstronomy simulations and scarce real world observations. The ultimate deliverable will be a robust, physically consistent, and highly interpretable deep learning pipeline capable of inferring the hidden properties of dark matter directly from noisy, real world astronomical data.

3 Technical Details

3.1 Gravitational Lensing Physics and Machine Learning Intersections

To effectively apply machine learning to astrophysics, the neural network must be grounded in the mathematics of General Relativity. In the strong lensing regime, the deflection of light is governed by the lens equation, which maps the true angular position of the background source, denoted as β, to the observed angular position of the image on the sky, denoted as θ. The lens equation is formally defined as $\beta = \theta - \alpha(\theta)$. The deflection angle, $\alpha(\theta)$, is determined by the gravitational potential of the foreground mass distribution.
Specifically, the deflection angle is the gradient of the two dimensional projected gravitational potential, scaled by cosmological distances. This is expressed as $\alpha(\theta) = \nabla\psi(\theta)$, where ψ is the scaled projected potential. The localized distortion and magnification of the background source are mathematically described by the Jacobian matrix of the lens mapping: $A = \frac{\partial\beta}{\partial\theta}$. The magnification μ at any point in the image plane is the inverse of the determinant of this Jacobian matrix: $\mu = \frac{1}{\det(A)}$.
When dark matter substructures such as low mass subhalos or axion vortex defects are present within the primary lensing galaxy, they introduce high frequency, localized perturbations into the continuous deflection field $\alpha(\theta)$. These microscopic perturbations alter the magnification and spatial positioning of the macroscopic Einstein rings or lensed arcs. The central objective of this project is to train a neural network to detect these minuscule spatial anomalies and classify the underlying dark matter model causing them.

3.2 Deficiencies of Purely Data Driven Deep Learning on Real Lensing

Images

Standard deep learning models, such as ResNet or Vision Transformer (ViT) architectures, operate as highly parameterized function approximators. When optimizing purely for categorical cross entropy on simulated datasets, these models excel by identifying highly specific pixel intensity textures associated with different dark matter classes. However, they remain entirely agnostic to the underlying lens equation or the Jacobian mapping.
When these purely data driven models are deployed on real lensing images, they encounter a severe domain shift. Real images contain Poisson noise inherent to photon counting, Gaussian read noise from Charge Coupled Devices (CCDs), and optical blurring mathematically represented by the telescope's Point Spread Function (PSF). Because standard CNNs have learned to associate subtle simulated pixel variations with dark matter substructure, they systematically misclassify real world noise artifacts as physical dark matter subhalos. Furthermore, without physical constraints, a CNN might predict a latent representation of a dark matter halo that, if fed back into the lens equation, would result in background source photons originating from entirely disparate, disconnected regions of the universe, a physical impossibility. To resolve this, the network's hypothesis space must be strictly constrained by the laws of physics

3.3 Core Methodology I: Lensing Constraints Embedded in Loss Functions

The first core methodology involves embedding explicit gravitational lensing constraints into the neural network's loss function. This transforms the architecture from a standard classifier into a true Physics Informed Neural Network (PINN)
My implementation will augment the standard supervised loss with a physics based regularization term. The total loss function will be structured as:
$\mathcal{L}_{total} = \mathcal{L}_{CE}(y, \hat{y}) + \lambda_{phys}\mathcal{L}_{physics}(\theta, \alpha)$ where $\mathcal{L}_{CE}$ is the standard cross entropy loss for the dark matter classification task, and λ_{phys} is a dynamic weighting coefficient that scales the influence of the physical constraint during training.

The physical loss term, $\mathcal{L}_{physics}$, will enforce the fundamental mapping of the lens equation. For a strongly lensed system featuring multiple images of a single background quasar or galaxy, all observed image positions θ_i must trace back to a singular, highly localized source position β_{source} . My architecture will include a differentiable module that outputs the predicted deflection field α_{pred} . The physical loss will compute the source plane positions for all detected image pixels: $\beta_i = \theta_i - \alpha_{pred}(\theta_i)$. The loss is then formulated as the spatial variance of these mapped source positions. If the network predicts an unphysical mass distribution that maps the observed arcs to widely separated regions of the source plane, the variance will be high, and the network will be heavily penalized. This explicitly forces the

network's latent space to learn the true physical parameters of gravitational deflection.

3.4 Core Methodology II: Analytical Mass Profile Priors (SIS and NFW)

To further constrain the network and prevent it from hallucinating arbitrary mass distributions, I am planning to integrate mass profile priors. Decades of astrophysical observation and N-body cosmological simulations have established that the mass density of galaxies follows highly specific mathematical profiles. My architecture will inject parameterized priors based on the Singular Isothermal Sphere (SIS) and the Navarro-Frenk-White (NFW) models.

The SIS profile perfectly models the combined baryonic and dark matter mass in the central regions of elliptical lensing galaxies. The three dimensional density is given by $\rho(r) = \frac{\sigma_v^2}{2\pi G r^2}$, where σ_v is the velocity dispersion. This yields a mathematically elegant, constant deflection angle. The NFW profile, conversely, models the extended dark matter halo and is defined by $\rho(r) = \frac{\rho_0}{(r/R_s)(1+r/R_s)^2}$, requiring specific scale radii (R_s) and concentration parameters (c).[27]

I will implement a dual stream architecture. While the primary classification stream analyzes the complex morphological features of the input image, a secondary continuous neural field stream will regress the macroscopic parameters of the SIS and NFW profiles (e.g., Einstein radius θ_E, concentration c, and velocity dispersion σ_v). By forcing the network's internal representation of the macroscopic lens to adhere to a linear combination of SIS and NFW analytical solutions, the primary classification stream is freed to focus purely on the micro perturbations caused by the dark matter substructure, dramatically improving signal to noise ratio and robustness on real data.

3.5 Core Methodology III: Bayesian Neural Networks for Uncertainty Quantification

A critical flaw of standard deep learning models is their inability to express "doubt." When presented with a heavily blurred HST image, a deterministic CNN will output a rigid, uncalibrated point estimate, often leading to catastrophic false positives in cosmological inference. To fix this I plan on transforming the physics guided architecture into a Bayesian Neural Network (BNN) to provide rigorous uncertainty quantification.

Instead of learning fixed scalar values for the convolutional weights, the BNN will learn probability distributions over the weights. By defining a prior distribution $P(\omega)$ over the weights ω, my objective is to compute the posterior distribution $P(\omega|\mathcal{D})$ given the real observational data $\mathcal{D}$. Because exact calculation of the posterior is computationally intractable for deep networks, I will utilize Variational Inference (VI).

The network will optimize the Evidence Lower Bound (ELBO):

L=−E[q(ω|θ)]+DKL(q(ω|θ) // P(ω)) where $q(\omega|\theta)$ is the variational distribution parameterized

by θ (mean and variance matrices), and D_{KL} represents the Kullback Leibler divergence acting as a regularization term. By utilizing the reparameterization trick, the network remains fully differentiable.

During inference on real lensing images, I will execute Monte Carlo (MC) forward passes, sampling different weight configurations from the learned posterior. The statistical variance in the output predictions across these multiple passes will mathematically quantify the epistemic (model) uncertainty. Consequently, if the network analyzes an ambiguous noise artifact, it will output a broad probability distribution rather than a sharp, false positive classification, ensuring that the pipeline produces statistically reliable science.

3.6 Supplementary Methodology I: Equivariant Neural Networks (ENNs)

To supplement the methodologies, I have extensively researched additional techniques to optimize the pipeline for real data. The first of these is the integration of Equivariant Neural Networks (ENNs). Strong gravitational lensing morphologies, particularly the subtle distortions caused by dark matter subhalos, exhibit continuous geometric symmetries. A dark matter substructure induces the exact same physical lensing perturbation regardless of its rotational orientation or spatial translation across the observer's sky.

Standard CNNs are translationally equivariant but completely lack rotational equivariance. When a standard CNN encounters a rotated Einstein ring, it must expend vast amounts of its parameter capacity to "re learn" the same physical features at the new angle. This is computationally inefficient and leads to severe overfitting on small datasets.

My architecture will incorporate $E(2)$ equivariant steerable convolutions. By mathematically defining the convolutional filters using circular harmonics, the network's feature maps are guaranteed to transform predictably under the action of the two dimensional Euclidean group $E(2)$ (encompassing translations, continuous rotations, and reflections). If an input lensing image is rotated by an arbitrary angle ϕ, the internal latent feature representations will rotate by the exact same group representation matrix. This geometric constraint drastically reduces the parameter count, prevents overfitting on orientation specific noise, and ensures that the model learns the true underlying physics of the dark matter morphology rather than orientation artifacts.

3.7 Supplementary Methodology II: Differentiable Ray Tracing Integration

A major limitation in current pipelines is the disconnect between the forward modeling simulation engines (e.g., Lenstronomy) and the backward pass optimization of the neural network. To bridge this gap, I will introduce Differentiable Ray Tracing directly into the PyTorch training loop.

In this paradigm, the neural network acts as a surrogate for the lens potential. The network processes the input image and outputs a continuous neural field representing the 2D gravitational potential $\psi(\theta)$ of the lensing galaxy. I will implement a PyTorch native differentiable renderer that takes this predicted potential, computes the deflection angle gradient $\nabla\psi$, warps a parameterized analytical background source profile, and renders a predicted observation on the image plane.
Because every tensor operation in this ray tracing sequence including grid generation and bilinear interpolation is fully tracked by PyTorch's autograd engine, the loss can be computed in pixel space between the rendered prediction and the actual observed HST/Euclid image. The gradients will flow seamlessly backward through the physical rendering equation directly into the neural network's weights.

3.8 Supplementary Methodology III: Multi Fidelity Learning Framework

The overarching hurdle in the project is the severe scarcity of high signal to noise, real world strong lensing images. Machine learning models require massive datasets to converge, but the astronomical community currently possesses only a limited catalog of verified strong lenses.
To solve this domain gap, I will implement a Multi Fidelity Learning framework. The PINN architecture will first undergo extensive pre training on a massive "low fidelity" dataset consisting of hundreds of thousands of rapidly generated, mathematically idealized Lenstronomy simulations. This phase teaches the network the fundamental principles of the lens equation and dark matter substructure interactions.
Following pre training, the network will be fine tuned on a targeted "high fidelity" dataset comprising scarce observational data from real telescopes and highly sophisticated simulations containing complex noise injection, CCD read errors, and atmospheric PSF blurring. To ensure seamless transfer, I will employ adversarial domain adaptation techniques, utilizing a discriminator network to force the PINN's latent feature space to become invariant to the source domain. This guarantees that the pristine physical relationships learned in simulation remain robust when exposed to the chaotic noise of the real universe.

3.9 Evaluation Task Results and Algorithmic Approach

To validate my capability to execute this complex proposal, I completed the official DeepLense evaluation tasks. The results unequivocally demonstrate my proficiency in deploying physics guided deep learning architectures, mirroring the requirements of the main GSoC 2026 project.

3.3.1 Common Test: Multi Class Classification

For the common evaluation task, the objective was to construct a robust deep learning pipeline capable of high confidence multiclass classification.

My approach utilized the EfficientNet B3 architecture, highly optimized for parameter efficiency, implemented strictly via PyTorch. I engineered a sophisticated data handling pipeline using torchvision.transforms.v2 to enforce robust spatial augmentations. More importantly, I utilized Sobel operators as an edge detection layer to distinguish between the smooth density perturbations of spherical subhalos and the sharp topological boundaries of Axion quantum vortices. This allowed the network to analyze the frequency domain spectral density and high frequency spatial gradients of the input arrays, extracting deep structural invariants rather than relying solely on spatial pixel intensities. This methodology yielded an exceptional consolidated **AUC-ROC Score of 98.59%**, proving the architecture's capacity to discriminate between subtle morphological classes with profound confidence.

3.3.2 Specific Test: Physics Guided Classification

For the specific evaluation task, I engineered and benchmarked a deep learning architecture designed for the high precision classification of strong gravitational lensing morphologies within the DeepLense framework. The core objective was to perform a comparative analysis between a deterministic convolutional baseline and a Physics Informed Neural Network (PINN) to quantify the performance gain achieved by embedding rigid physical inductive biases directly into the optimization pipeline.

Utilizing the PyTorch framework, I adapted a ResNet18 backbone to ingest and process simulated lensing arrays. My methodology involved a two stage training protocol: first establishing a performance floor via a purely data driven supervised baseline, followed by the deployment of a physics guided variant. This PINN architecture replaces standard pixel agnostic feature extraction with an optimization process that strictly enforces the underlying gravitational mechanics governing spacetime distortions.

To rigorously validate the efficacy of these models, I utilized the Area Under the Receiver Operating Characteristic (AUC-ROC) metric. The physics informed architecture demonstrated superior discriminative power, achieving a consolidated **AUC-ROC Score of 87.73%**, significantly eclipsing the baseline performance of **84.39%**. This delta unequivocally validates that the integration of physical constraints into deep learning architectures dramatically enhances predictive fidelity, generalization, and scientific robustness when operating on complex cosmological datasets.

4 Project Deliverables

4.1 Objective and Implementation Details

The primary objective of this GSoC project is to deliver a fully operational, open source Physics Informed Neural Network (PINN) pipeline, entirely native to PyTorch, specifically tailored for the extraction of dark matter morphology from real gravitational lensing observations.

The core deliverables to the ML4SCI repository will include:

- **Multi Fidelity Dataset Curation Pipeline:** Modular Python scripts to aggregate, clean, and preprocess real observational data (HST/Euclid) alongside high fidelity Lenstronomy simulations.
- **Mass Profile Priors:** The deployment of the core model, featuring E(2) Equivariant convolutional layers, Bayesian Variational Inference capabilities, and dual stream mass profile prior networks enforcing SIS/NFW geometries.
- **Differentiable Ray Tracing Engine:** A PyTorch based differentiable rendering module enabling continuous loss computation.
- **Evaluation and Benchmarking Suite:** Comprehensive metrics calculators (AUC-ROC for classification, Log Likelihood for Bayesian uncertainty calibration, and physical residual tracking) comparing the PINN approach against baseline ResNet and ViT models.
- **Extensive Scientific Documentation:** Highly detailed Jupyter notebooks, comprehensive inline PEP8 code documentation, and a final academic research report detailing the methodologies and physical justifications.

4.2 Timeline for Deliverables

To accommodate the integration of all six advanced physical and mathematical methodologies, I have structured an intensive, highly detailed 12 week timeline.

Phase	Timeframe	Key Milestones and Technical Objectives
Pre GSoC	May 4 - May 25	**Community Bonding & Mathematical Formulation:** Finalize the exact tensor calculus derivations for the $\mathcal{L}_{physics}$ loss functions with mentors. Review the DeepLense codebase and study foundational literature on Bayesian Neural Networks and Differentiable Ray Tracing.
Phase 1	Week 1	**Multi Fidelity Dataset Architecture:** Curate the "low fidelity" simulated dataset utilizing Lenstronomy, generating varied CDM, WDM, and Axion substructure morphologies. Aggregate the "high fidelity" real lensing dataset, applying standard astronomical preprocessing (PSF deconvolution, background subtraction). Develop optimized PyTorch DataLoaders.
	Week 2	**Baseline Implementations:** Develop a standard Vision Transformer (ViT) and ResNet50 baseline on the real data to establish a performance floor. Implement automated evaluation scripts (AUC-ROC, Confusion Matrices). *Deliverable: Initial GitHub PR with baseline metrics.*
	Week 3	**Lensing Constraints in Loss Functions:** Code the custom $\mathcal{L}_{physics}$ loss term that calculates the variance of source plane mappings $\beta = \theta - \nabla\Psi_{pred}$. Execute hyperparameter sweeps on the penalty weight λ_{phys}.

	Week 4	**Differentiable Ray Tracing Integration:** Build the PyTorch native differentiable ray tracing module to replace analytical approximations. Enable end to end backpropagation from the rendered image plane to the latent space. Verify gradient flow through the physical parameters.
Phase 2	Week 5	**Mass Profile Priors (SIS & NFW):** Create the secondary neural stream designed to regress the parameters of the SIS and NFW profiles. Couple these outputs into the Differentiable Ray Tracer to render the prior constrained base image.
	Week 6	**Mid Term Evaluation:** Conduct comprehensive training runs combining the lensing constraints, mass priors, and ray tracing. Document the performance delta against Week 2 baselines. Draft a comprehensive blog post detailing the physics informed mechanisms. *Deliverable: Mid term evaluation report*.
	Week 7	**Transition to Bayesian Neural Networks (BNNs):** Convert the deterministic convolutional layers into Bayesian layers via Variational Inference. Code the Evidence Lower Bound (ELBO) loss function, integrating the Kullback-Leibler (KL) divergence.
	Week 8	**Uncertainty Quantification Calibration:** Develop the Monte Carlo forward pass inference scripts. Extract epistemic uncertainty maps for real lensing images. Calibrate the output probabilities to ensure the network expresses appropriate doubt when encountering high noise real images.
Phase 3	Week 9	**Equivariant Neural Networks (ENNs):** Refactor the network backbone to utilize E(2) Steerable Convolutions. Define the circular harmonic group representations to ensure the network is strictly invariant to the rotation and translation of dark matter subhalos across the observer's sky.
	Week 10	**Multi Fidelity Transfer Learning:** Execute the final training regime. Pre-train the entire physics informed, Bayesian, Equivariant architecture heavily on the simulated low fidelity dataset. Unfreeze terminal layers and fine tune on the high fidelity real observational dataset using domain adaptation techniques.
	Week 11	**Benchmarking and Code Refinement:** Run exhaustive hyperparameter tuning. Utilize Mixed Precision (FP16) and torch.compile to optimize inference latency and memory consumption. Generate final ROC curves and uncertainty density maps. Refactor codebase to adhere to strict PEP8 standards.
	Week 12	**Final Deliverables & Submission:** Submit finalized pull requests to the DeepLense repository. Publish the final blog post

		summarizing the journey from simulated constraints to real world Bayesian inference. Discuss post GSoC integration strategies with mentors. *Deliverable: Final codebase and comprehensive research report*.

5 Other Information

5.1 Why ML4SCI?

The intersection of fundamental cosmology and state of the art deep learning represents the absolute frontier of modern scientific discovery. The ML4SCI organization, with its direct operational ties to high energy physics, CERN, and advanced cosmological research, provides an unparalleled environment to apply computational mechanics to the universe's most profound mysteries. I am deeply motivated by the philosophical premise that deep learning should not merely function as a statistical pattern matcher, but must be mathematically constrained to uncover invariant physical truths. The DeepLense project perfectly aligns with my long term academic ambition to transition from traditional engineering into a PhD program focused exclusively on Scientific Machine Learning (SciML) and computational astrophysics. Contributing to an open source framework that actively aids humanity's search for dark matter is an exceptionally compelling and personal pursuit.

5.2 Academic Background

I am currently pursuing a Bachelor of Engineering (B.E.) in Mechanical Engineering at the Birla Institute of Technology and Science (BITS Pilani), Goa Campus, maintaining excellent academic standing. This rigorous engineering curriculum has provided me with a formidable mathematical foundation in partial differential equations, calculus, linear algebra, and complex 3D modeling all of which are directly analogous to the mathematics of gravitational potentials and neural network gradients. To pivot this traditional engineering background into computational science, I have heavily augmented my university coursework with specialized technical certifications, including comprehensive, rigorous training in Supervised Machine Learning (Regression and Classification), Advanced Learning Algorithms, and Unsupervised Learning.

5.3 Past Experience

I possess a robust portfolio of technical projects that demonstrate a high degree of proficiency in deep learning architecture, physical simulation, and systems engineering.

- **DeepLense Evaluation Tasks:** I independently developed advanced ML models for physics guided classification, achieving exceptional AUC-ROC scores (98.59% on the common benchmark and 87.73% on the physics specific task). This required building custom PyTorch CNN and ResNet architectures from scratch, implementing spectral data augmentations, and executing complex physics informed feature extraction protocols.

- **SEDS Celestia (CubeSat: Project Antara):** Serving as a core engineering team member (2025–2026), I contribute heavily to the design and development of a CubeSat tasked with analyzing atmospheric radiation within the South Atlantic Anomaly (SAA). This role requires complex 3D modeling and strict physical simulation using SolidWorks and Fusion 360, bridging the gap between theoretical physics and applied engineering.
- **Cloud Architecture & Hackathons:** I have demonstrated strong cloud deployment capabilities under pressure, winning "Best Use of Linode Cloud" and "Best Use of Google Cloud" at separate international hackathons, ensuring I can deploy and scale the heavy computing required for astronomical machine learning.

5.4 Past Interaction with Mentors

I have proactively engaged with the ML4SCI community and the specific mentors assigned to this project to ensure my proposal perfectly aligns with the organization's scientific objectives. Through detailed discussions with **Pranath Reddy**, I shared my evaluation task results. He reviewed my metrics, and provided essential encouragement to proceed with formulating this proposal.
Furthermore, invaluable theoretical guidance was provided by **Prof. Prasanta Kumar Das**, an astrophysics professor at BITS Pilani. Upon reviewing my initial project approach, Prof. Das explicitly recommended the integration of three advanced techniques to solve the real data bottleneck: embedding lensing constraints directly within the loss function, enforcing mass profile priors (SIS/NFW) to regularize predictions, and utilizing Bayesian approaches for rigorous uncertainty quantification to capture degeneracies in lensing solutions. These specific, highly technical mentor recommendations have been deeply woven into the core methodology of this proposal, guaranteeing that the project will deliver maximum scientific value to the DeepLense collaboration.

5.5 Commitments

My academic schedule at BITS Pilani is highly flexible during the summer months. If selected for GSoC 2026, I will dedicate a full time effort (minimum 35-40 hours per week) exclusively to this project. All extraneous academic and extracurricular responsibilities have been cleared for the duration of the GSoC coding period to ensure complete, uninterrupted focus on delivering a high quality, production ready codebase. I am fully available for daily asynchronous communication and weekly synchronous meetings via Slack, Zoom, or Teams in the Indian Standard Time Zone (IST) GMT+5:30. Should any unavoidable circumstances arise, I will communicate proactively with my mentors and ensure all development milestones are met precisely on schedule.

6 Post GSoC Improvements

My genuine passion for this research drives my ambition to integrate generative models and universal representations, ensuring that our work not only stays at the absolute frontier of scientific machine learning but also actively shapes the way we interpret the next generation of astronomical data.

1. Physics Informed Diffusion Models for Dataset Synthesis Traditional generative Adversarial Networks (GANs) often produce statistically plausible but physically unconstrained outputs, which is detrimental for creating high fidelity gravitational lensing datasets. To address this, I propose implementing Latent Diffusion Models (LDM) augmented with strict physics informed priors. This approach utilizes a score based generative process: a forward pass corrupts real lensing images with noise, and a modified U-Net learns the reverse denoising process to synthesize new data. To ensure scientific viability, I will incorporate differentiable physics constraints (e.g., consistency with the lens equation) and mass profile priors (SIS/NFW) directly into the LDM's loss function. This hybrid architecture would enable conditional generation of complex substructure morphologies, dramatically accelerating the creation of scientifically consistent training data for the DeepLense pipeline.

2. Hybrid Quantum Classical Representation Learning As the high energy physics and astrophysics communities inevitably transition toward the Quantum computing era, traditional classical tensor operations will face insurmountable computational bottlenecks when simulating the incredibly complex, micro scale interactions of dark matter substructures. I plan to explore Hybrid Quantum Classical Representation Learning for DeepLense. By mathematically encoding the highly dimensional gravitational lensing parameter spaces into quantum states via Parameterized Quantum Circuits (PQCs), and seamlessly combining these with classical Bayesian neural networks (e.g., using Quantum Model Agnostic Meta Learning frameworks), the DeepLense pipeline could theoretically process immense volumes of astronomical data exponentially faster. This would exploit quantum entanglement to perfectly model the complex, non linear dependencies between baryonic matter, dark matter subhalos, and the differential deflection of light across the cosmos.

References:

1. Physics Guided Machine Learning on Real Lensing Images - Archive Project Details | Google Summer of Code https://summerofcode.withgoogle.com/archive/2025/projects/lfQvKrOC
2. Enhancing Gravitational Lens Study with Deep Learning: A Study on Effects of Dropout Regularization - MDPI https://www.mdpi.com/2075-4434/14/2/18
3. Equivariant Neural Networks for Signatures of Dark Matter Morphology in Strong Lensing Data - Machine Learning and the Physical Sciencehttps://ml4physicalsciences.github.io/2023/files/NeurIPS_ML4PS_2023_188.pdf
4. Equivariant Neural Networks for Dark Matter Morphology with Strong Gravitational Lensinghttps://ml4sci.org/gsoc/2022/proposal_DEEPLENSE3.html
5. Lens-JEPA: Physics Informed Joint Embedding Predictive Architecture for Gravitational Lensing https://ml4physicalsciences.github.io/2025/files/NeurIPS_ML4PS_2025_340.pdf
6. Image Simulations for Strong and Weak Gravitational Lensing - MDPI

https://www.mdpi.com/2073-8994/12/4/494
7. Physics Informed learning on Real Gravitational Lensing | by Ashutosh Ojha | Medium https://medium.com/@ojhaaashutosh1005/physics-informed-learning-on-real-gravitational-lensing-5bb8f1d8248b
8. Equivariant Neural Networks for Signatures of Dark Matter Morphology in Strong Lensing Data - NeurIPS https://neurips.cc/virtual/2023/76224
9. Equivariant Neural Fields for Dark Matter Morphology Classification and Other Applications - Vishal Chandra https://vishalchandra.com/data/ENFsGravReport.pdf
10. Extended ray-mapping method based on differentiable ray-tracing for non-paraxial and off-axis freeform illumination lens design - Optics Letters https://opg.optica.org/oe/abstract.cfm?uri=oe-31-19-30066
11. Image Based Heliostat-Calibration in Solar Tower Power Plants with Differentiable Kinematics and Artificial Intelligence Bildbas https://elib.dlr.de/218951/1/Master_Thesis___Tristan_Wortberg.pdf
12. A Graphical Multi-Fidelity Gaussian Process Model, with Application to Emulation of Heavy-Ion Collisions | Request PDF - ResearchGate https://www.researchgate.net/publication/375531885_A_graphical_multi-fidelity_Gaussian_process_model_with_application_to_emulation_of_heavy-ion_collisions
13. Conglomerate Multi-fidelity Gaussian Process Modeling, with Application to Heavy-Ion Collisions | Request PDF - ResearchGate https://www.researchgate.net/publication/381001787_Conglomerate_Multi-fidelity_Gaussian_Process_Modeling_with_Application_to_Heavy-Ion_Collisions
14. The Carousel Lens II: Cosmological Constraints with GIGA-Lens - arXiv https://arxiv.org/pdf/2602.16077
15. GSOC 25 with ML4SCI | Foundation Model for Gravitational Lensing | by Rishirswamy https://medium.com/@rishirswamy/foundation-model-for-gravitational-lensing-f6e75ab24d88
16. MIU2Net: weak-lensing mass inversion using deep learning with nested U-structuresAll python code and data that we used to generate the figures in the paper are available at https://github.com/MIU2NET/miu2net. - arXiv https://arxiv.org/html/2601.13538v1
17. Sensing prior constraints in deep neural networks for solving exploration geophysical problems - PMC https://pmc.ncbi.nlm.nih.gov/articles/PMC10265955/
18. Review of Physics-Informed Neural Networks: Challenges in Loss Function Design and Geometric Integration - MDPI https://www.mdpi.com/2227-7390/13/20/3289
19. LensPINN: Physics Informed Neural Network for Learning Dark Matter Morphology in Lensing - NeurIPS 2026 https://neurips.cc/virtual/2024/100100
20. Perturbed f (R) gravity coupled with neutrinos: Exploring cosmological implications | Request PDF - ResearchGate https://www.researchgate.net/publication/388150140_Perturbed_f_R_gravity_cou

pled_with_neutrinos_Exploring_cosmological_implications
21. publications | François Lanusse https://flanusse.net/publications/
22. arXiv:1909.02005v2 [astro-ph.CO] 17 Oct 2019 https://arxiv.org/pdf/1909.02005
23. Galaxy mass modelling from multiwavelength JWST strong lens analysis: dark matter substructure, angular mass complexity, or both? | Monthly Notices of the Royal Astronomical Society | Oxford Academic https://academic.oup.com/mnras/article/539/2/704/8099938
24. Strong Lensing by Galaxies - PMC - NIH https://pmc.ncbi.nlm.nih.gov/articles/PMC11549236/
25. Strong Lensing by Galaxies - arXiv https://arxiv.org/html/2210.10790v4
26. The Wendelstein Weak Lensing (WWL) pathfinder: accurate weak lensing masses for Planck clusters - Oxford Academic https://academic.oup.com/mnras/article-pdf/486/1/77/28498997/stz817.pdf
27. Physics-Informed Neural Networks for Modeling Galactic Gravitational Potentials - arXiv https://arxiv.org/html/2602.01806v1
28. Constraining cosmological parameters from N-body simulations with variational Bayesian neural networks - Frontiers https://www.frontiersin.org/journals/astronomy-and-space-sciences/articles/10.3389/fspas.2023.1139120/full
29. Hierarchical Inference with Bayesian Neural Networks: An Application to Strong Gravitational Lensing (Journal Article) - OSTI https://www.osti.gov/biblio/1772050
30. Large-scale Gravitational Lens Modeling with Bayesian Neural Networks for Accurate and Precise Inference of the Hubble Constant (Journal Article) | OSTI.GOV https://www.osti.gov/biblio/1773678
31. [1708.08843] Uncertainties in Parameters Estimated with Neural Networks: Application to Strong Gravitational Lensing - arXiv https://arxiv.org/abs/1708.08843
32. Scalable Bayesian Inference for Finding Strong Gravitational Lenses - Machine Learning and the Physical Sciences https://ml4physicalsciences.github.io/2022/files/NeurIPS_ML4PS_2022_155.pdf
33. Classifying modified gravity and dark energy theories with Bayesian neural networks: massive neutrinos, baryonic feedback, and the theoretical error | Monthly Notices of the Royal Astronomical Society | Oxford Academic https://academic.oup.com/mnras/article/535/4/3141/7881578
34. Machine Learning and the Physical Sciences, NeurIPS 2025 https://ml4physicalsciences.github.io/
35. Differentiable strong lensing: uniting gravity and neural nets through differentiable probabilistic programming | Monthly Notices of the Royal Astronomical Society | Oxford Academic https://academic.oup.com/mnras/article/496/1/381/5848214
36. Computational Wavefront Sensing: Theory, Practice, and Applications https://vccimaging.org/Publications/Wang2021Dissertation/Wang2021Dissertation.pdf

37. Lensformer: A Physics-Informed Vision Transformer for Gravitational Lensing https://ml4physicalsciences.github.io/2023/files/NeurIPS_ML4PS_2023_214.pdf
38. Open Research Online https://oro.open.ac.uk/91821/1/Andrew_Davies_PhD_Thesis_Changes.pdf
39. Multifidelity Deep Operator Networks For Data-Driven and Physics-Informed Problems | Request PDF - ResearchGate https://www.researchgate.net/publication/373620270_Multifidelity_Deep_Operator_Networks_For_Data-Driven_and_Physics-Informed_Problems
40. UC San Diego Electronic Theses and Dissertations - eScholarship.org https://escholarship.org/content/qt5d86g8wg/qt5d86g8wg.pdf
41. Opportunities in AI/ML for the Rubin LSST Dark Energy Science Collaboration FERMILAB-PUB-25-0886-CSAID-PPD arXiv:2601.14235https://lss.fnal.gov/archive/2025/pub/fermilab-pub-25-0886-csaid-ppd.pdf